ENGLISH VIDEO RESOURCES FOR TEACHING STEM SUBJECTS

Tài nguyên video tiếng Anh cho giảng dạy các môn STEM

Maths • Physics • Chemistry • Biology • Engineering • IT

Use note / Lưu ý sử dụng: Link to or embed videos from the original platform rather than re-uploading copies unless you have permission. Teachers can use the videos for pre-teaching, flipped learning, vocabulary support, concept explanation, or follow-up discussion. Nên liên kết hoặc nhúng video từ nền tảng gốc thay vì tải xuống và đăng lại nếu chưa có quyền. Video có thể dùng để dạy trước, flipped learning, hỗ trợ từ vựng, giải thích khái niệm hoặc thảo luận sau khi xem.

1. MATHS / TOÁN

Topic / Nội dung	Suggested level / Khối gợi ý	Video / Resource	Key vocabulary / Từ khóa	Teaching idea / Gợi ý sử dụng
Linear Equations / Phương trình bậc nhất	Grades 8–10	**Khan Academy – Linear Equations 1** Open video/resource	equation, variable, coefficient, solve	Pause before each solution step and ask students to predict the next operation. / Dừng trước mỗi bước và yêu cầu học sinh dự đoán phép biến đổi tiếp theo.
Two-variable Linear Equations / Phương trình hai biến	Grades 8–10	**Khan Academy – Graphing Lines** Open video/resource	slope, intercept, x-axis, y-axis	Ask students to label the graph in English and explain slope using a sentence frame.
Geometry – Triangles / Hình học – Tam giác	Grades 7–10	**Khan Academy – Geometry: Triangles** Open video/resource	angle, side, vertex, triangle, congruent	Pre-teach 5 geometry words, then let students identify them while watching.
Probability / Xác suất	Grades 7–11	**Khan Academy – Introduction to Probability** Open video/resource	outcome, event, sample space, probability	Use coin or dice examples after the video and ask students to describe outcomes in English.
Statistics / Thống kê	Grades 9–12	**Khan Academy – Statistical Studies** Open video/resource	population, sample, survey, experiment	Have students distinguish population and sample using one local classroom example.

2. PHYSICS / VẬT LÝ

Topic / Nội dung	Suggested level / Khối gợi ý	Video / Resource	Key vocabulary / Từ khóa	Teaching idea / Gợi ý sử dụng
Forces / Lực	Grades 8–10	**Khan Academy – Forces** Open video/resource	force, contact force, net force, balanced	Students create a two-column vocabulary table while watching.
Newton's Laws / Các định luật Newton	Grades 9–11	**Khan Academy – Newton's Laws of Motion** Open video/resource	inertia, acceleration, mass, action, reaction	After watching, students explain one real-life example using: "According to Newton's ___ Law…"
Motion & Forces / Chuyển động và lực	Grades 9–12	**Khan Academy – High School Physics** Open video/resource	velocity, acceleration, displacement, force	Use as a unit resource library; assign one short video before class.
Electric Circuits / Mạch điện	Grades 8–11	**Khan Academy – Intro to Electric Circuits** Open video/resource	circuit, battery, resistor, current, voltage	Students draw the circuit while listening and label each component in English.
Energy / Năng lượng	Grades 9–12	**Khan Academy – High School Physics: Energy** Open video/resource	energy, work, kinetic, potential, conservation	Ask students to make a concept map after the selected energy video.

3. CHEMISTRY / HÓA HỌC

Topic / Nội dung	Suggested level / Khối gợi ý	Video / Resource	Key vocabulary / Từ khóa	Teaching idea / Gợi ý sử dụng
Atomic Structure / Cấu tạo nguyên tử	Grades 8–10	**Khan Academy – Atomic Structure and Properties** Open video/resource	atom, proton, neutron, electron, nucleus	Students complete a labelled atom diagram while watching.
Periodic Table / Bảng tuần hoàn	Grades 8–10	**Khan Academy – The Periodic Table** Open video/resource	group, period, metal, non-metal, valence	Pause to ask: "What do elements in the same group have in common?"
Chemical Reactions / Phản ứng hóa học	Grades 8–11	**Khan Academy – Chemical Reactions** Open video/resource	reactant, product, reaction, evidence	Students identify reactants/products from a simple equation after watching.
Balancing Equations / Cân bằng phương trình	Grades 9–11	**Khan Academy – Types of Chemical Reactions** Open video/resource	coefficient, equation, balance, atom	Use one worked example, then let students explain why atoms must be conserved.
Acids and Bases / Axit và bazơ	Grades 9–12	**Khan Academy – Acids, Bases and pH** Open video/resource	acid, base, pH, neutral, indicator	Students classify household examples and use: "___ is acidic/basic because…"

4. BIOLOGY / SINH HỌC

Topic / Nội dung	Suggested level / Khối gợi ý	Video / Resource	Key vocabulary / Từ khóa	Teaching idea / Gợi ý sử dụng
Cells / Tế bào	Grades 7–10	**Khan Academy – Structure of a Cell** Open video/resource	cell, organelle, nucleus, membrane, cytoplasm	Students label a cell diagram before and after watching to compare learning.
Prokaryotic & Eukaryotic Cells / Tế bào nhân sơ và nhân thực	Grades 9–11	**Khan Academy – Prokaryotic and Eukaryotic Cells** Open video/resource	prokaryote, eukaryote, nucleus, DNA	Students complete a compare/contrast table in English.
Photosynthesis / Quang hợp	Grades 8–11	**Khan Academy – Photosynthesis** Open video/resource	chloroplast, light energy, glucose, oxygen	Use a process diagram and sentence frame: "Photosynthesis is the process by which…"
DNA / ADN	Grades 9–12	**Khan Academy – DNA** Open video/resource	DNA, nucleotide, base pair, helix, gene	Students note 5 key terms and then explain DNA structure to a partner.
Ecology / Sinh thái học	Grades 8–12	**Khan Academy – Ecology** Open video/resource	ecosystem, population, food web, biodiversity	Use one local ecosystem example after the video to activate student discussion.

5. ENGINEERING / KỸ THUẬT

Topic / Nội dung	Suggested level / Khối gợi ý	Video / Resource	Key vocabulary / Từ khóa	Teaching idea / Gợi ý sử dụng
Introduction to Engineering / Giới thiệu kỹ thuật	Grades 8–12	**Crash Course Engineering – Full Course** Open video/resource	design, system, constraint, prototype, engineer	Use selected episodes to introduce different engineering fields.
Engineering Playlist / Danh sách video kỹ thuật	Grades 9–12	**Crash Course Engineering – YouTube Playlist** Open video/resource	civil, mechanical, electrical, materials	Assign different teams one episode and ask them to summarise the engineering field.
Engineering Design Process / Quy trình thiết kế kỹ thuật	Grades 7–12	**MIT K-12 – Engineering Design Process** Open video/resource	define, design, build, test, improve	Students use the five verbs as headings in their own design project.
Engineering Process for Younger Learners / Quy trình kỹ thuật cơ bản	Grades 6–9	**Crash Course Kids – Engineering Process** Open video/resource	problem, solution, test, redesign	Good pre-task video before a bridge or tower challenge.
Bridges & Structures / Cầu và kết cấu	Grades 8–12	**Practical Engineering – Every Kind of Bridge Explained**	beam, arch, truss, suspension, load	Students identify bridge types and explain one advantage of each.

		Open video/resource		
Materials / Vật liệu kỹ thuật	Grades 10–12	**Crash Course – Materials, Stresses & Toughness** Open video/resource	stress, strain, toughness, elasticity, fatigue	Use with a material-comparison table or simple design challenge.

6. IT / COMPUTER SCIENCE / TIN HỌC

Topic / Nội dung	Suggested level / Khối gợi ý	Video / Resource	Key vocabulary / Từ khóa	Teaching idea / Gợi ý sử dụng
Computer Basics / Kiến thức máy tính cơ bản	Grades 6–10	**Code.org – How Computers Work** Open video/resource	computer, input, output, memory, processor	Select short videos and have students complete a hardware/software sorting task.
Binary & Data / Nhị phân và dữ liệu	Grades 7–11	**Code.org – Binary & Data** Open video/resource	binary, bit, byte, data, representation	Students encode small numbers or letters after the video.
Internet & Networks / Internet và mạng	Grades 7–12	**Code.org – How the Internet Works** Open video/resource	IP address, DNS, packet, routing, HTTP	Assign one video per concept and create a class network glossary.
Cybersecurity / An ninh mạng	Grades 8–12	**Code.org – Cybersecurity & Crime** Open video/resource	encryption, security, password, attack, privacy	Use a short scenario discussion after watching.
Algorithms & Scratch / Thuật toán và Scratch	Grades 8–12	**CS50 – Introduction to Computer Science** Open video/resource	algorithm, pseudocode, input, output, loop	Use the algorithms section and ask students to write pseudocode for a familiar task.
Algorithms – Advanced / Thuật toán nâng cao	Grades 10–12	**CS50 – Algorithms** Open video/resource	linear search, binary search, sorting, recursion	Compare two algorithms using efficiency and step count.

Suggested Website Structure / Gợi ý cấu trúc trên website

Subject → Topic → Watch Video → Key Vocabulary → Teaching Task

Ví dụ: Physics → Newton's Laws → Watch video → force, mass, acceleration → students explain one real-life example using a sentence frame.

7. ADDITIONAL VIDEO SOURCES / NGUỒN VIDEO BỔ SUNG

Purpose / Mục đích: These additional sources broaden the resource library beyond Khan Academy, Crash Course, Code.org and CS50. They can be used for lesson preparation, flipped learning, vocabulary support, enrichment, demonstrations and project-based learning. / Các nguồn bổ sung này giúp mở rộng thư viện ngoài Khan Academy, Crash Course, Code.org và CS50; có thể dùng cho chuẩn bị bài, flipped learning, hỗ trợ từ vựng, mở rộng kiến thức, minh họa và học theo dự án.

Maths / Toán

Topic / Nội dung	Suggested level / Khối gợi ý	Video / Resource	Key vocabulary / Từ khóa	Teaching idea / Gợi ý sử dụng
Algebra, Geometry, Probability, Trigonometry	Grades 7–11	**FuseSchool – Maths video library** Open video/resource	algebra, graph, angle, probability, trigonometry	Use short clips for pre-teaching and ask students to note 3–5 key maths terms in English.
Calculus / Giải tích	Grades 11–12 / Extension	**MIT OpenCourseWare – Introductory Science and Math** Open video/resource	derivative, integral, function, rate of change	Use selected lecture segments for advanced students or teacher enrichment.
Statistics & Graphing / Thống kê và đồ thị	Grades 10–12	**Bozeman Science** Open video/resource	mean, distribution, graph, variable, trend	Pause at graph examples and ask students to describe the trend in English.

Physics / Vật lý

Topic / Nội dung	Suggested level / Khối gợi ý	Video / Resource	Key vocabulary / Từ khóa	Teaching idea / Gợi ý sử dụng
Forces, Motion, Energy, Waves, Electricity	Grades 8–11	**FuseSchool – Physics** Open video/resource	force, motion, energy, wave, current	Good for short EMI clips; pair with a bilingual vocabulary list.
Mechanics, Energy, Electricity / Cơ học, năng lượng, điện	Grades 10–12	**Bozeman Science – Physics** Open video/resource	velocity, acceleration, energy, field, circuit	Use one concept video before problem-solving practice.
Microgravity, Mass, Motion / Vi trọng lực, khối lượng, chuyển động	Grades 8–12	**NASA STEMonstrations** Open video/resource	mass, center of mass, rotation, gravity	Excellent real-world context; ask students to predict outcomes before watching.
Classical Mechanics; Electricity & Magnetism	Grades 11–12 / Extension	**MIT OpenCourseWare** Open video/resource	mechanics, field, charge, magnetism	Use for teacher enrichment or selected advanced excerpts.

Chemistry / Hóa học

Topic / Nội dung	Suggested level / Khối gợi ý	Video / Resource	Key vocabulary / Từ khóa	Teaching idea / Gợi ý sử dụng
Matter, Reactions, Acids & Bases	Grades 8–11	**FuseSchool – Chemistry** Open video/resource	matter, reaction, acid, base, particle	Use clips as pre-lab preparation and vocabulary support.
General & AP Chemistry	Grades 10–12	**Bozeman Science – Chemistry** Open video/resource	bonding, equilibrium, kinetics, stoichiometry	Assign one video as homework and discuss key terms in class.
Atomic Theory, Periodic Trends, Bonding	Grades 11–12 / Extension	**MIT OpenCourseWare – Principles of Chemical Science** Open video/resource	atomic theory, periodicity, ionic bond, molecular structure	Use selected segments for extension or teacher professional learning.

Biology / Sinh học

Topic / Nội dung	Suggested level / Khối gợi ý	Video / Resource	Key vocabulary / Từ khóa	Teaching idea / Gợi ý sử dụng
Cells, Genetics, Plants, Physiology, Ecology	Grades 8–11	**FuseSchool – Biology** Open video/resource	cell, gene, organ, ecosystem, adaptation	Use a short clip then ask students to complete a concept map.
Biology, Genetics, Ecology, Physiology	Grades 10–12	**Bozeman Science – Biology** Open video/resource	DNA, inheritance, homeostasis, ecology	Useful for flipped learning and review before assessment.
Biodiversity, Ecology, Conservation	Grades 7–12	**National Geographic Education** Open video/resource	biodiversity, habitat, conservation, ecosystem	Use for discussion tasks and local-global comparison activities.
Fundamentals of Biology	Grades 11–12 / Extension	**MIT OpenCourseWare** Open video/resource	cell biology, genetics, evolution, physiology	Use short lecture segments for advanced learners or

				teacher enrichment.

Engineering / Kỹ thuật

Topic / Nội dung	Suggested level / Khối gợi ý	Video / Resource	Key vocabulary / Từ khóa	Teaching idea / Gợi ý sử dụng
Engineering Design, 3D Printing, Space Engineering	Grades 7–12	**NASA STEM / NASA eClips** Open video/resource	design, prototype, 3D printing, system, test	Use before design challenges; students identify the design problem and constraints.
Materials, Mechanical, Electrical Engineering	Grades 10–12 / Extension	**MIT OpenCourseWare** Open video/resource	materials, mechanics, electrical, system, design	Use selected segments for enrichment and teacher background knowledge.
Real-world Civil Engineering	Grades 8–12	**Practical Engineering** Open video/resource	structure, foundation, load, bridge, infrastructure	Ask students to connect video examples with local infrastructure.
Engineering Disciplines & Systems	Grades 9–12	**Crash Course Engineering** Open video/resource	civil, mechanical, electrical, systems, design	Assign different episodes to groups and compare engineering fields.

IT / Computer Science / Tin học

Topic / Nội dung	Suggested level / Khối gợi ý	Video / Resource	Key vocabulary / Từ khóa	Teaching idea / Gợi ý sử dụng
Computer Science, Algorithms, Discrete Maths	Grades 10–12 / Extension	**MIT OpenCourseWare – Mathematics for Computer Science** Open video/resource	algorithm, graph, logic, probability, proof	Use selected segments for advanced learners or teacher enrichment.
Programming & Computer Science	Grades 9–12	**freeCodeCamp YouTube** Open video/resource	programming, variable, function, data structure, algorithm	Choose short sections from long tutorials and provide a task checklist.
Technology, Computing & Digital Literacy	Grades 7–12	**TED-Ed Technology videos** Open video/resource	data, internet, AI, computing, security	Use as discussion starters with comprehension questions.
Internet, Networks, Cybersecurity	Grades 7–12	**Code.org Video Library** Open video/resource	network, packet, DNS, encryption, cybersecurity	Use one short video per concept and build a class glossary.

Recommended order for Vietnamese secondary teachers / Thứ tự ưu tiên gợi ý: FuseSchool → Bozeman Science → NASA STEM → National Geographic Education → Khan Academy → MIT OpenCourseWare. MIT resources are best placed under “Advanced / Extension Resources”.